ONYX · AGENTIC INTELLIGENCE

Agentic AI Implementation & Market Strategy

Exactly what is implemented, what is deterministic or simulated, and how XELOR can turn governed manufacturing intelligence into a defensible product advantage.

Prepared from source code, module contracts, database controls, the supplied managed-services deck and primary standards

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1. The honest answer

XELOR has implemented meaningful agentic infrastructure—not merely a chat box. The application contains a nine-agent registry, versioned bounded graphs, live tenant-scoped ERP and platform capability calls, parallel execution waves, durable run/node state, checkpoints, evidence, human approval pauses, approval ancestry enforcement, action dispatch records and an operator-facing Mission Control. RELAY provides the managed-service operating layer; ACHILES adds private hourly platform checks and immutable health history without receiving any repair or ERP-write authority.

What is real now: orchestration, authorization, ERP reads, persistence, checkpoints, tenant-scoped pending approvals, global pending count, consequence review, mandatory decision notes, Approve/Reject, audit trails, governance switches, deterministic evaluation, Copilot queries and governed internal action dispatch.

What is not active in the current demo configuration: no hosted LLM is configured; Agent OS language reasoning is deterministic; there are zero live external connectors; a dispatched action creates an internal governed work item and does not autonomously post inventory, money or supplier messages. Railway packaging exists, but a live hosted URL is not proven by the repository.

Why this is a good foundation

- Models are outside the transaction and permission boundaries.
- Capabilities are a closed allow-list, not arbitrary SQL or APIs.
- Side effects require a successful human-approval ancestor.
- Provider outage degrades to deterministic behavior instead of blocking ERP.
- Every model call and consequential agent act is attributable and auditable.

What must improve

- Connect and benchmark at least one production model provider.
- Add production-grade retrieval, document ingestion and semantic authorization.
- Expand feature-specific golden sets and online evaluation.
- Connect verified external signals through governed adapters.
- Connect approved actions to domain executors only after shadow-mode evidence.

LIVE real runtime/data path **DETERMINISTIC** code-generated reasoning/fallback **CURATED DEMO** clearly labelled scenario data **FUTURE** recommended production capability.

2. AI and agentic features implemented today

Capability	Implementation	Current truth	Control boundary
Shared AI router	One `AiRouterService`: validate registry → governance check → provider → hash-chained log → usage ledger	LIVE	Unknown/unroutable feature keys fail closed.
Provider abstraction	`AiProvider` contract with deterministic stub and optional Ollama adapter	LIVE STUB DEFAULT	Provider selection is configuration; feature code does not receive provider credentials.
AI governance	Tenant opt-out, per-feature/global kill switches, daily token budget, usage recording	LIVE	All shared-router calls meet the same database-backed gate.
AI action log	Per-tenant monotonic append-only SHA-256 hash chain; input/output content hashes, model, tier, usage and degraded state	LIVE	Raw prompt/PII is not placed in the AI action chain.
Feature portfolio	Canonical 8-feature portfolio plus an explicit in-evaluation Copilot entry and an Integration null entry	LIVE	Status/risk/data class/degraded mode are code-owned.
Evaluation harness	Golden sets, baseline vs candidate, F1/macro-F1, must-pass rules and exit-code ship gate	LIVE	A candidate must outperform the deterministic baseline and pass hard cases.
Grounding gate	Feature-specific checks that reject invented identifiers, reversed conclusions and unsupported narration	LIVE	Code owns decisions; the model may phrase accepted evidence.
ONYX Copilot	Closed intent router, permission check, hand-written SELECT, template answer, source citation, optional narration	LIVE	Read-only by construction; no generated SQL and no business-write path.
Agent OS	Versioned graphs, agents, live capability calls, waves, checkpoints, approvals, verification, cancellation and durable history	LIVE REASONING	Capabilities re-check user RBAC and agent allow-lists.
Decision intelligence	Cross-module risk rules, evidence graph, governed recovery start and estimated-versus-verified outcome ledger	LIVE DETERMINISTIC	Five ERP domains are checked; missing monetary rates remain blank instead of being invented.
Human approval inbox	Global top-bar count, dedicated tenant-scoped queue, exact consequence preview, required note, Approve/Reject and Mission Control handoff	LIVE	Only users with `agentos.approval.decide` see decision controls; server checks the permission again.
Working Capital intelligence	RASP cash-position, forecast simulation, collections and finance-pack capabilities plus a bounded Working Capital Review graph	LIVE CONTROLS CURATED UI KPIs	RASP cannot move money; simulations do not change invoices, vouchers, payments or bank instructions.
QMS & Audit intelligence	KILN inspection/evidence reads, audit/CAPA/pack drafts plus a bounded QMS & Audit Readiness graph	LIVE CONTROLS CURATED WIDER QMS	KILN cannot declare compliance, confirm root cause, release documents or close CAPA.
Controlled actions	Six approved `governed_work_item` dispatches persisted in an append-only ledger	LIVE INTERNAL	Engine rejects side effects without an approved ancestor.
Signal ingress	Typed local ERP signal starts an idempotent controlled mission	LIVE LOCAL	`eventId` is the replay boundary; no external webhook is claimed.
Copilot rail extras	Alerts, morning brief and action recommendations	CURATED DEMO	Clearly labelled; review links do not claim execution.

3. The nine-agent operating model

ONYX
Supervisor; frames missions, delegates, joins evidence and publishes outcomes.

HEXA
Governance; verifies boundaries, evidence, permissions and side-effect ancestry.

MICA
Sales & Product Care; reads customer commitments, explains manufactured-product after-sales cases and proposes governed follow-up work.

SPAR
Supply; reads stock and proposes supply-chain actions.

AXLE
Planning; reads planned orders and proposes planning actions.

KILN
Operations & Quality; reads production, inspection and stored Factory Connect robot/AMR/dwell records, then prepares QMS, audit and controlled plant work.

RASP
Finance & Working Capital; reads finance evidence, simulates cash scenarios and prepares controlled work.

RELAY
Managed Services; owns the service clock, hand-offs, customer updates, change calendar and improvement while specialists retain diagnosis and repair.

ACHILES
Private Platform Assurance; checks availability and freshness, records evidence and hands failure to RELAY and the technical owner.

Controlled mission sequence

ONYX bounds goal → 8 registered reads parallel wave → 8 specialist assessments

ONYX joins plan → HEXA preflight → Human approval durable pause

7 approved work items → HEXA verifies approval ancestry → ONYX publishes outcome

Human-in-the-loop control as implemented

Mission pauses at approval node → Pending count visible globally → Human opens approval inbox

Review exact consequence → Write reason + Approve / Reject → Resume or cancel with audit event

- **The agent cannot self-approve.** The approval node remains waiting until a separately authorised user submits a decision.
- **Authority is narrow.** The person approves the exact proposed action shown in the inbox, not a broad permission grant.
- **The reason is evidence.** A decision note is mandatory and stored with approver identity and decision time.

- **Rejection is real.** Rejecting cancels the mission; approval resumes the persisted graph and does not skip verification.
- **Discovery is product-wide.** The top-bar Approvals control is visible from every workspace and updates when a mission reaches or leaves a human gate.

Three bounded specialist missions

Mission	Specialist evidence	AI contribution	Human-only decisions
Working Capital Review	RASP finance position and forecast simulation, MICA customer commitments, SPAR stock position.	Explain risks, separate facts from assumptions, compare safe options and draft a concise ONYX brief.	Customer contact, supplier timing, payment, financing submission, bank instruction and ledger posting.
QMS & Audit Readiness	KILN inspection records, traceable evidence collection and draft evidence-pack manifest.	Find relevant proof, mark gaps, suggest investigation questions and draft readiness language.	Compliance verdict, document release, audit conclusion, confirmed root cause, product disposition and CAPA closure.
Factory Flow Recovery	KILN reads stored robot/AMR/dwell records; MICA, SPAR, AXLE and RASP add business consequences.	Explain the bottleneck and check the declared boundary. When a simulator command is requested, the mission freezes its exact typed intent for human review; a separate API call may later evaluate that same intent against stored state.	The graph has no dispatch node. The API records only a simulated policy result: no raw motion, program upload, safety decision, edge delivery or controller acknowledgement is claimed.

Why this qualifies as agentic

- The system decomposes a bounded objective across specialized identities and executes independent work concurrently.
- It persists state across steps, waits for external human input, resumes, branches/verifies and records evidence.
- Agents use registered capabilities, not direct database access, and their authority is narrower than the application’s.
- Actions are gated on runtime-verified approval ancestry rather than trusting the graph author or model.

> **Important nuance:** “agentic” describes the orchestration and controlled capability model. It does not mean the current demo uses free-form LLM reasoning at every node.

3A. Agent-to-module contracts in detail

An agent's business ownership is broader than its callable runtime surface. The modules below explain the context the specialist understands; the capability column is the smaller set of data it can actually fetch during a mission.

Agent	Modules it represents	Direct mission evidence	What its conclusion must preserve
ONYX	ONYX Copilot; Agent OS; AI Operations.	Company context only through `general.companies.read`; all other evidence arrives from specialists.	Scope, source links, parallel results, HEXA verdict, approval state and unresolved limits. ONYX cannot dispatch an action.
HEXA	Organisation; Administration; Integration.	Company masters through `general.companies.read`; graph/user/tool/audit state through the run context.	Verified tenant and actor, permission intersection, policy checks, approval ancestry and auditability. Simulated connector health cannot be labelled externally verified.
MICA	Sales; Customer Care & Warranty.	Tenant-scoped sales commitments through `sales.orders.read`.	Customer/order/date/credit evidence and the difference between promised, dispatched and still open. Product-case, warranty, spares or feedback context cannot be invented when it was not retrieved. Operational problems with XELOR itself belong to RELAY.
SPAR	Purchase; Inventory.	Non-zero on-hand balances through `inventory.on-hand.read`.	Item, warehouse, batch and available quantity as observed. A purchase expectation is not on-hand stock, and an approved PO is not a receipt.
AXLE	Engineering; Planning.	Latest planned orders and status/source filters through `planning.planned-orders.read`.	Proposal, quantity, due/release dates, source and plan limitation. It must not present infinite-capacity MRP or a draft schedule as feasible execution.
KILN	Production; QMS & Audit; Maintenance.	Production orders, inspections, quality evidence, Factory Connect robot/AMR/dwell views and draft audit/CAPA/pack artifacts.	Observed evidence and draft labels. It cannot decide product release, compliance, root cause, CAPA effectiveness or local machine safety.
RASP	Accounts; Employee Spend; People; Working Capital.	Vouchers, trial balance, bounded scenario evidence, collection review and draft finance-pack manifest.	Posted fact versus scenario assumption versus illustrative screen value. It cannot move money, contact a customer, certify a funding pack or post a voucher.
RELAY	Managed Services.	Service catalogue, lifecycle, incidents, changes, reviews and the responsibility map through `managed-services.service-assurance.read`.	Service coordination only. Technical repair, security determination, AI controls, factory assets, product warranty and contracts stay with their existing owners.
ACHILES	Private Platform Assurance.	Current platform status, component checks, latency and immutable history through `platform-health.status.read`.	Detection only. It cannot diagnose, restart, repair, dispatch work or communicate with customers.

> **Why this separation matters:** deep module knowledge helps a specialist explain consequences, but runtime data access remains intentionally narrow. Adding a module to an agent's description never grants a database read; a registered capability and the requesting person's permission are both still required.

3B. Exact capability surface and modes

The registry contains 19 unique capabilities. Eighteen are read, analyse, simulate or draft. The only effectful capability is shared by seven specialists and remains structurally downstream of human approval; ACHILES is excluded from it.

Agent	Callable capabilities	Mode and evidence	Approval/effect boundary
ONYX	`general.companies.read`	Read company/registration context.	No execute capability; supervisor is intentionally least powerful.
HEXA	`general.companies.read` `agent.action.dispatch`	Read context; append governance work item.	Dispatch requires `agentos.run.operate` and approved ancestor.
MICA	`sales.orders.read` `agent.action.dispatch`	Read commitments; append commercial work item.	No customer contact, order override or shipment.
SPAR	`inventory.on-hand.read` `agent.action.dispatch`	Read balances; append supply work item.	No PO, receipt or stock movement.
AXLE	`planning.planned-orders.read` `agent.action.dispatch`	Read current plan; append planning work item.	No firming, conversion, MRP rerun or schedule publication.
KILN	`production.factory-connect.read`; `production.orders.read`; `quality.inspections.read`; `quality.evidence.collect`; `quality.audit-plan.draft`; `quality.capa-plan.draft`; `quality.audit-pack.draft`; `agent.action.dispatch`	Read stored robot/AMR/dwell and ERP evidence, analyse and prepare labelled drafts. Its dispatch capability belongs to the separate controlled-action graph; Factory Flow itself dispatches nothing.	No raw machine command, local-safety verdict, inspection verdict, disposition, document release, audit conclusion, CAPA closure or production posting.
RASP	`accounts.vouchers.read`; `finance.cash-position.read`; `finance.forecast.simulate`; `finance.collections.prioritise`; `finance.funding-pack.draft`; `agent.action.dispatch`	Read posted evidence; return bounded scenario/review/draft; append approved finance work.	No payment, bank instruction, posting, customer message or certification.
RELAY	`managed-services.service-assurance.read`; `agent.action.dispatch`	Read service evidence; append one approved coordination item.	No technical repair, security/compliance decision, AI-control change, contract or service credit.
ACHILES	`platform-health.status.read`	Read private status, freshness and result history.	No dispatch, diagnosis, service restart, ERP write or customer communication.

What the single effect actually writes

1. The graph engine proves that an approval node is an ancestor of the dispatch node and that its recorded decision is yes.
2. The capability repeats the agent allow-list and requesting-user permission checks.
3. An idempotent, append-only `governed_work_item` is written with run, node, agent, approval and payload references.
4. The mission later verifies that the work items exist. It does not claim the underlying departmental work is complete.

> **Common demo misunderstanding:** “seven actions dispatched” means six domain assignments plus one RELAY service-coordination assignment were created after approval. It does not mean XELOR paid a supplier, changed stock, contacted a customer, altered a plan or operated a live service desk.

3C. Coordination, deployment and identity modes

How one mission is coordinated

Person / signal fixed tenant + actor → ONYX starts versioned graph → Specialists read parallel wave

Domain services enforce rules → HEXA verifies joined evidence → Human approves or rejects

Approved work items only → Checkpoint + audit persist outcome → ONYX publishes limits included

Runtime mode	Identity source	Agent controls retained	Allowed use
Normal local / future production	Keycloak OIDC Authorization Code + PKCE; API verifies signature, issuer, expiry and groups.	Tenant context, user permissions, agent allow-list, graph bounds, approval ancestry and RLS.	Development now; production only after the wider security/operations roadmap.
Railway public demo	Explicit Web/API demo flags map a small fixed persona list to the isolated Northstar tenant; web sends the dedicated demo marker.	Capability permissions, agent allow-list, graph/approval checks and RLS still run against the mapped persona.	Disposable investor data only. Never a shortcut for a pilot or real tenant.

Hosted demo service boundary

Next.js web public → NestJS API private → PostgreSQL + worker private durable state

Railway service files start Web, API, Worker, PostgreSQL and Valkey; API boot waits for the database, applies migrations, seeds/checkpoints the demo version once and then reports readiness. The agent architecture is unchanged by hosting: the deterministic stub remains the provider, and no external connector is made real merely because the URL is public.

> **Do not infer deployment from files:** the repository proves repeatable packaging and local parity. It does not prove an active Railway project, a public domain, TLS, monitoring or availability.

3D. RELAY managed-service operating model

RELAY is the human-service coordination layer around XELOR. It follows the service pattern visible in the supplied Cisco CoSol deck—design, transition, operate and improve—adapted to XELOR rather than copied as a network-service claim.

Design catalogue + SLO → **Transition** discover + accept → **Operate** incident + change → **Improve** review + actions

Task	Accountable owner	RELAY hand-off and hard boundary
Service catalogue, onboarding, incident clock, customer change calendar, SLO evidence and service review	**RELAY**	Owns coordination and customer communication; cannot claim a technical fix without specialist evidence.
Connector failure, retry, DLQ, mapping, security determination and audit	**HEXA**	RELAY links customer impact; HEXA alone diagnoses the connector and decides security/reportability.
Provider, prompt, evaluation, guardrail, autonomy and kill switch	**ONYX AI Operations**	RELAY may coordinate service impact; it cannot change any AI control.
Manufactured-product tickets, warranty, AMC and complaints	**MICA**	No duplicate Managed Services ticket unless the XELOR technology service itself is affected.
Factory machine, PM, downtime and maintenance work order	**KILN**	Physical restoration remains KILN; a machine breakdown is not automatically a XELOR service incident.
SLA credit, contract, scope and material customer promise	**Authorised human**	RELAY assembles evidence; no agent signs or grants credit.

Dedicated assurance graph

ONYX frames outcome → **RELAY** reads + assesses → **HEXA** verifies boundary → **Human gate** RELAY brief

> **Honest demo boundary:** incidents, changes, reviews and service measures are seeded illustrative operating-model data. The repository does not prove a staffed 24×7 desk, live ITSM/telemetry, customer paging or a contractual SLA.

Design basis: ISO/IEC 20000-1; ITIL incident/problem/request/change/service-level practices; NIST SP 800-61 Rev. 3; Google SRE SLO/error-budget guidance; OpenTelemetry.

4. ONYX Copilot: how it is actually safe

User question → Closed intent classification → Permission check

Hand-written tenant query → Template answer from rows → Citation + question log

Six controls in the current implementation

1. **Understand deterministically first.** A model may only rescue an unresolved question by choosing from the closed intent catalogue.
2. **Authorize twice.** The user must hold Copilot permission and the permission required by the equivalent business screen.
3. **Retrieve with registered SQL.** Queries are hand-written, row-capped and executed under the caller's tenant transaction.
4. **Render from evidence.** Numeric answers come from returned rows, not model memory.
5. **Narrate only when opted in.** Narration is off by default; unsupported numbers or claims are rejected and the template answer remains.
6. **Log every outcome.** Answered, clarified, refused and forbidden questions record intent, confidence, sources and row count.

Strong design choice

A hosted model can improve language understanding without receiving arbitrary database access. The retrieval surface remains a finite catalogue owned by product code.

Current limitation

The Copilot answers exactly 21 pre-registered intents. This makes it safe and predictable, but it is not yet a broad conversational analyst over every workflow or document.

Recommended evolution

Keep the same authorization/query boundary. Add semantic intent routing, query composition from approved metrics, authorized document retrieval and multi-turn context gradually; never replace tenant/RBAC filters with prompt instructions.

5. Current AI portfolio and maturity

Feature	Risk posture	Implemented foundation	Next production step
Receipt/document extraction + categorisation	Draft record; PII-minimised	Extraction service, arithmetic re-derivation and eval specification	Vision/document model pilot, field confidence, review UI, dataset by vendor/layout.
Master-data duplicate detection	Draft record; business data	Deterministic detection, code-owned verdict, optional explanation and grounding gate	Shadow-test model explanations; add feedback and multilingual/vendor-name cases.
Ticket triage + sentiment	Advisory	Classifier, golden-set macro-F1 gate, human acceptance distinction	Run shadow mode, monitor per-class recall—especially statutory/rare categories.
Duplicate receipt detection	Advisory	Exact hash baseline and registered stretch slot	Add perceptual/document similarity only after labelled fraud/duplicate cases.
HSN/SAC suggestion	Draft record	Registered stretch slot and deterministic directory baseline	Use approved tax catalogue retrieval, confidence thresholds and mandatory review.
Reply drafting/thread summary	Draft only	Draft service and rule gate against promises/verdicts	Provider pilot with customer tone library, evidence links and send-time human approval.
Payslip explainer	Advisory only forever	Template/explainer path	Ground every sentence to payslip/rate-book lines; never expose cross-employee data.
SoD conflict explanation	Advisory only forever	Rule-owned conflict with explanation service	Use model only to explain the rule and remediation choices; never grant/revoke access.
Copilot retrieval Q&A	In evaluation; read only	Closed intents, registered queries, permissions, citations and refusal	Hosted semantic router, authorized RAG, live operational feed and online quality metrics.

> **Registry governance issue to formalize:** the code describes a canonical closed set of eight but also contains an explicit ninth in-evaluation Copilot feature. That divergence is documented in source; production governance should require a signed ADR and versioned registry change process.

6. What is simulated versus real in the demo

Demo surface	Real	Simulated/curated	How to present it
Authenticated gateway	Keycloak login; catalogue call; 9/9 registry status	Visual topology/animation	“The topology is visual; the connection status is read from the runtime catalogue.”
Mission Control	Graph start, ERP reads, state, evidence, checkpoint, approval, resume and dispatch ledger	Language summaries are deterministic	“The operating controls are live; model reasoning is not enabled in this environment.”
Copilot chat	Permissions, tenant query, template answer, citation and log	Optional model routing/narration inactive by default	“Answers come from governed queries; the model is not allowed to invent SQL.”
Copilot Alerts/Brief/Actions tabs	Open the current Decision Commander and live Human Approval inbox	No permanent scenario cards remain in these tabs	Fiction is isolated inside an explicitly active guided demo.
Guided demo case	Navigates one canonical seeded Northstar commitment across nine evidence-linked acts and clears instantly through the top Stop demo control	The guide layer is presentation-only; every named business fact is read from persisted ERP records	A compact lower-right guide explains each act while leaving the live ERP screen unobstructed and visually unchanged. The complete journey has automated read-only coverage and makes no hidden writes.
Controlled actions	Append-only approved internal work-item dispatches	No external supplier/API/financial posting	“Approval created accountable domain work; external execution is the next integration phase.”

Never claim: that eight independent LLMs are currently running; that Agent OS autonomously changed ERP ledgers; that RELAY is a staffed 24×7 service; that external systems were contacted; or that every AI feature uses a production model.

7. Recommended production AI architecture

Product UI / ERP event → Intent + policy gateway → Agent graph runtime

Authorized retrieval SQL + documents → Provider router small/large/vision → Structured result schema + grounding

Human approval when consequential → Capability broker domain services → Audit + eval + telemetry loop

Recommended stack additions

Layer	Recommendation	Reason
Hosted inference	Add one adapter first: OpenAI Responses API or Amazon Bedrock Converse/Responses access; preserve `AiProvider`	Structured tools, streaming and production capacity without coupling domains to a vendor.
Model portfolio	Small/fast classifier-router; stronger reasoning model for bounded analysis; vision model for documents/inspection images; embedding model for retrieval	Route by task, risk, latency and cost rather than using one expensive model everywhere.
Retrieval	PostgreSQL + pgvector first; hybrid lexical/vector search; reranking; document chunks with tenant/role/effectivity metadata	Uses existing database and keeps authorization close to source records.
Document pipeline	S3/KMS, malware scan, OCR/layout extraction, chunking, embedding, version/effectivity and retention	Turns SOPs, manuals, certificates and drawings into governed evidence.
Prompt/config registry	Versioned prompt packs and schemas with author/reviewer, model settings and rollback	Makes AI releases inspectable and independently approvable.
Guardrails	Existing PII scan + provider guardrails/moderation; prompt-injection checks; output schemas; citation/number validation	Defense in depth; provider controls supplement but do not replace application authorization.
Evaluation	Keep custom deterministic gates; add provider eval jobs, adversarial sets, SME review and online outcome metrics	Measures business correctness, not just fluent text.
Observability	OpenTelemetry traces with model, feature, latency, usage, retrieval ids, refusal/degradation—without raw sensitive payloads	Links AI behavior to the same operating view as APIs, queues and approvals.
Feedback	Accept/edit/reject reasons, final human outcome, correction categories and dataset curation workflow	Creates the proprietary learning loop competitors cannot buy from a model vendor.

8. Roadmap to an outstanding market position

Phase A — Safe intelligence pilot (0–90 days)

1. Implement a hosted provider adapter and structured output contract; retain deterministic fallback.
2. Enable model use only for Copilot intent rescue, master explanation and ticket triage in shadow mode.
3. Expand golden sets with real Indian manufacturing language, abbreviations, mixed English/Hindi/Tamil and rare compliance categories.
4. Add feature/model/prompt version, latency, cost, refusal and human disposition dashboards.
5. Complete privacy impact assessment, retention controls, provider review and red-team prompt injection.

Phase B — Evidence-grounded assistants (3–6 months)

1. Build authorized ingestion for SOPs, machine manuals, inspection standards, drawings and certificates.
2. Add plant morning brief from live events, not curated cards: exceptions, evidence, owner, due date and link.
3. Launch draft-only document and reply features: receipt extraction, customer reply, RCA evidence pack and supplier follow-up.
4. Introduce a governed metric/semantic catalogue so Copilot can answer cross-module questions without generated SQL.

Phase C — Controlled action (6–12 months)

1. Connect `governed_work_item` dispatches to domain-specific executors one at a time.
2. Start with reversible/low-risk drafts: purchase requisition draft, maintenance task draft, reschedule proposal, ticket assignment.
3. Require diff previews, policy simulation, named approver, idempotency, postcondition verification and automatic rollback/compensation where possible.
4. Keep inventory posting, payroll, GL posting, statutory submission and supplier payment human-authorized with stricter multi-person controls.

Phase D — Manufacturing intelligence moat (12–24 months)

1. Cross-order risk graph linking demand, material, capacity, quality, maintenance, cash and workforce evidence.
2. Predictive maintenance and quality anomaly models trained on customer-consented operational data.
3. What-if planning copilot that compares scenarios using deterministic solvers and uses the LLM only to explain trade-offs.
4. Benchmark network and anonymized priors only with explicit governance; tenant data never silently trains another tenant’s experience.

9. Highest-value differentiated AI use cases

Use case	Why customers pay	Correct technical pattern	Risk limit
Decision & risk commander	Explains delivery, supply, planning, quality and maintenance risks, connected value, causes and recovery choices.	Live deterministic cross-module analysis + persisted evidence links + governed mission start.	Live recommendations and approval creation; domain execution remains approval-bound.
Planner scenario copilot	Turns “what if supplier X slips 7 days?” into schedule, inventory and cash consequences.	MRP/solver owns numbers; model parses request and explains scenario deltas.	Never let the LLM calculate the production plan itself.
Quality investigation assistant	Builds traceable NCR/8D evidence packs from inspection, batch, machine, supplier and complaint records.	Authorized retrieval + structured RCA hypotheses + human confirmation.	Hypotheses are labelled; disposition remains a controlled decision.
Maintenance reliability advisor	Prioritizes likely failure risk and parts/work windows around production commitments.	Statistical/anomaly model + asset history + approved planning capability.	No automatic machine shutdown from an LLM recommendation.
Document-to-transaction drafting	Reduces manual entry of receipts, invoices, certificates and orders.	Vision extraction → schema → deterministic totals/tax/duplicate checks → review.	Draft only; confidence/field provenance visible.
Commercial response assistant	Produces accurate order/status replies backed by live evidence.	Template facts + optional controlled narration + send approval.	No promise, price, date or concession without policy/human approval.
Compliance control explainer	Makes SoD, GST, payroll, audit and security exceptions understandable.	Rule engine decides; model explains the rule, source and remediation path.	Never let the model reinterpret statutory calculation or grant access.

Competitive thesis: XELOR should not compete on “having a chatbot.” Its moat can be a verified manufacturing evidence graph plus safe action infrastructure: every answer tied to current plant records, every recommendation tested against deterministic rules, and every consequential act attributable.

10. Evaluation and safety system

Release ladder for every AI feature

Offline golden set baseline comparison → **Grounding + adversarial gate** → **Shadow mode no user effect**

Small pilot human review → **Limited release kill switch** → **Continuous outcome monitor**

Metric family	Measure	Examples
Task correctness	Feature-specific result quality against labelled truth	Macro-F1 per ticket category; field accuracy; retrieval recall; grounded numeric accuracy.
Safety	Forbidden action/data exposure and unsupported claim rate	Cross-tenant leak = zero; unapproved write = zero; invented identifiers = zero.
Human outcome	What operators accepted, edited, rejected and eventually completed	Draft acceptance, edit distance, recommendation conversion, false-alert burden.
Operations	Latency, availability, degradation and budget	p50/p95 response; provider failure; fallback rate; tokens and ₹ per successful outcome.
Fairness/coverage	Quality by language, role, site, category and data sparsity	Rare statutory tickets, mixed-language questions, smaller plants and new tenants.
Drift	Input/retrieval/outcome change over time	New terminology, vendor layouts, seasonal plan patterns and model-version regressions.

Non-negotiable safety invariants

- Tenant and role authorization is enforced in code/database, never delegated to a prompt.
- No raw database credentials, generated SQL or broad HTTP tool are exposed to a model.
- Every tool has a schema, owner, allow-list, idempotency policy and side-effect classification.
- Consequential execution has a preview, attributable approval and verified postcondition.
- Provider failure, refusal or budget exhaustion must leave core ERP workflows available.
- Private chain-of-thought is neither requested nor stored; structured evidence and decision summaries are enough.

11. Build, buy and framework choices

Decision	Recommendation	Reasoning
Agent orchestration	Keep XELOR's graph engine now; evaluate Temporal as the durability substrate when workflow complexity demands it	The current engine already encodes domain authorization and approval ancestry. Replacing it with a generic agent library would risk weakening the important part.
Model provider	Buy inference; keep a thin multi-provider adapter	Foundation-model training is not XELOR's moat. Evaluation data, manufacturing semantics and controlled tools are.
Vector store	Use pgvector first	It is already present and can keep tenant/effectivity metadata near relational authorization.
RAG framework	Implement a small explicit ingestion/retrieval service before adopting a broad framework	Authorization, citations, version/effectivity and evals matter more than framework convenience.
Guardrails	Combine XELOR checks with provider guardrails; do not outsource policy	Provider filters can detect PII/prompt attacks; business permissions and transaction safety remain application responsibilities.
Evaluation platform	Retain code-level ship gates; optionally export runs to managed eval/observability tools	Critical must-pass cases should remain reproducible in CI and independent of one vendor console.
Fine-tuning	Defer	Prompt/retrieval/eval quality and proprietary labelled feedback should mature first; tune only for a measured gap.

> **Principle:** buy commodity model capability; build the manufacturing truth layer, permissioned capabilities, evaluation corpus and operator feedback loop.

12. Prioritized implementation backlog

Priority	Deliverable	Proof of completion
P0	Hosted provider adapter with structured outputs, timeout/retry/circuit breaker and deterministic fallback	Same feature contract passes stub, hosted and provider-failure tests.
P0	AI telemetry and prompt/model/version registry	Every call has correlation, feature, version, latency, usage, result and degradation without raw sensitive payload.
P0	Registry governance ADR for Copilot addition	Versioned portfolio change approved by product, security and domain owner.
P0	Expanded red-team and golden sets	Prompt injection, tenant/role probes, rare classes and multilingual questions pass hard gates.
Delivered foundation	Decision Commander and verified-outcome ledger	Current cross-module risks link to source evidence; marketing value is not counted as verified value.
P1	Authorized document ingestion/RAG	Retrieval respects tenant/role/effectivity; source recall and citation accuracy meet threshold.
P1	Human feedback learning workflow	The live approval inbox already captures attributable reasons; next, curate privacy-reviewed accept/edit/reject outcomes into labelled evaluation cases.
P1	First real controlled executor	A reversible draft action runs only after approval and passes replay/postcondition tests.
P2	Scenario/planning assistant	Deterministic engine reproduces plan; model explanation contains no unsupported numbers.
P2	Quality and maintenance intelligence pilots	Prospective shadow evaluation beats existing rules without increasing missed critical events.

Board-level success measures

- Minutes saved per approved workflow, not tokens consumed or chats started.
- Fewer late orders, expedites, repeated defects and unplanned downtime attributable to earlier evidence.
- Draft acceptance with low edit distance and no increase in control exceptions.
- Zero tenant leaks, zero unapproved consequential actions and fast kill-switch recovery.
- Gross margin contribution after inference/review/operations cost.

13. Evidence and official references

Repository evidence inspected

- `apps/api/src/ai/*` — router, governance, providers, action log, feature services and eval harness
- `packages/platform/src/ai/*` and `packages/platform/src/aiops/*` — registry, evaluation, privacy and routing primitives
- `apps/api/src/modules/copilot/*` — intent, permission, retrieval, render, narration and question log
- `apps/api/src/agent-os/*` and `packages/platform/src/agent-os/*` — graph runtime, capabilities, repository, controlled actions and authorization
- `packages/db/src/schema/agent-os.ts` and Agent OS migrations
- `apps/web/src/spine/shell/copilot-rail.tsx`, global Human Approvals link, ONYX gateway, approval inbox and Mission Control screens
- Working Capital and QMS & Audit module manifests, RASP/KILN capability registrations and bounded graph definitions
- All 24 web module manifests and their owning domain controllers/services, including RELAY's managed-service operating model and ACHILES platform-health API, screen, scheduler and append-only evidence table
- `infra/railway` public-demo identity, startup, health and five-service deployment files
- Agent OS Phase 1–3 documentation and authenticated browser tests, including start → discover → note → approve → resume

Official references informing the future architecture

- OpenAI function/tool calling and JSON-schema tools: developers.openai.com/api/docs/guides/function-calling
- OpenAI evaluation guidance/API: developers.openai.com/api/docs/guides/evals
- OpenAI platform data controls: developers.openai.com/api/docs/guides/your-data
- Amazon Bedrock Guardrails: docs.aws.amazon.com/bedrock/latest/userguide/guardrails.html
- Amazon Bedrock evaluations: docs.aws.amazon.com/bedrock/latest/userguide/evaluation.html
- OpenTelemetry JavaScript: opentelemetry.io/docs/languages/js/
- ISO/IEC 20000-1 service management: iso.org/standard/70636.html
- NIST incident response: NIST SP 800-61 Rev. 3
- Google SRE service-level objectives: sre.google/sre-book/service-level-objectives/

Final strategic recommendation

XELOR should become the manufacturing system that can explain every recommendation, prove every source, and safely turn approved decisions into work. Keep deterministic engines responsible for money, stock, tax, scheduling constraints and control verdicts. Use models for language, extraction, classification, evidence synthesis and hypothesis generation—and let the existing Agent OS make their use bounded, observable and accountable.

This report describes the working repository inspected on 8 August 2026. It deliberately distinguishes live orchestration from deterministic reasoning and RELAY's illustrative operating model from a staffed managed service. Future provider/product choices require security, privacy, contractual and jurisdiction-specific review.